

From the labs

Close to hitting the market, we explore the phenomenon of Modular smartphones.

**Samsung leads in India**

With a 17% increase in unit shipments in Q216, Samsung still leads the Indian smartphone market <http://digit.in/SmsngLeads>

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Space facts that sound like BS...

... but are actually true

Space is an unforgiving place, inhabited by objects that push the limits of physics and human imagination. Even against that background, here are ten facts about space and our universe that at first will still sound like bullsh*t but are not.

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01 One of the largest known structures in our cosmos is a hole and it's so huge that you can fit 10,000 galaxies inside it

It is not surprising to read about a stellar body that is much larger than the Earth. In fact, what do you define as huge anymore? Jupiter is huge. A galaxy is huge, so much so that a mere planet is like a grain of sand in comparison. How about a hole as big as 10,000 galaxies? Scientists have reported the existence of a void in space that was missing about 10,000 galaxies. It has been pegged as one of the largest individual structures ever identified by humankind. Christened as a 'supervoid' this 'hole' in space is not a vacuum but simply a region in space that is extremely barren, cold and devoid of mass. To put a number to how large the void is, imagine that you could travel at the speed of light (it's blasphemy to even imagine that though) and wished to traverse the diameter of the hole. It would take you around 1,400 million years. To put things further into perspective, the largest man-made hole on Earth is in Utah, USA and it's about four kilometers wide. The supervoid is roughly 10^{14} times larger!

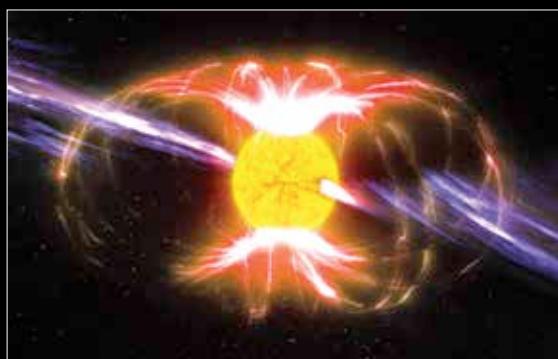
02 There is a cloud made of alcohol in space and no you cannot drink from it

While at first glance this might sound to be truly BS, if you think about it, alcohol is a fairly simple compound. It is made up of Carbon, Hydrogen and Oxygen and all three exist aplenty in space. The complex formation processes and chemistry that is found in interstellar gas clouds could be the birthplace of an alcohol cloud formation. And a process called quantum tunneling can, in theory, explain how alcohol is formed even in the harshness of space. Alright, all good in theory, but did we actually find one? As it turns out we've found plenty of such clouds. One such formation has 10 billion-billion-billion litres of space booze. Let's not even try to comprehend that number. Another cloud formation is 'just' 6,500 light years away. In cosmic scales this particular cloud region named W3(OH) is practically our neighbour. So given a large enough straw, or a fast enough rocket, could we tap into the seemingly endless supply of alcohol? Unfortunately not. Most of the alcohol is

methanol which we don't consume for fear of death. There are traces of ethyl alcohol, but peppered between other toxic compounds. Ahh, space cocktail!

03 There are magnets in space that are powerful enough to break every chemical bond that exists in our body - from a thousand kilometres away

At the end of a star's dying stages there sometimes occurs an explosion dubbed as a supernova, which under the right conditions results in the formation of a neutron star. Neutron Stars are very dense objects packed with neutrons and some protons. More importantly, neutron



An artist's impression of a magnetar showing radio emissions and magnetic field (CC 3.0)

stars also rotate really fast. Closely packed charged particles rotating at an insane pace produce strong magnetic fields. A certain class of neutron stars, dubbed magnetars have magnetic fields so powerful that they would make our Earth bound magnets seem like kiddie toys. The most powerful magnet on